

Pascal's Hyper-Pyramids



CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

4.2

Medium

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[Southwestern Europe
Regional Contest \(SWERC\)
2016](#)



Pascal's triangle of height 5:

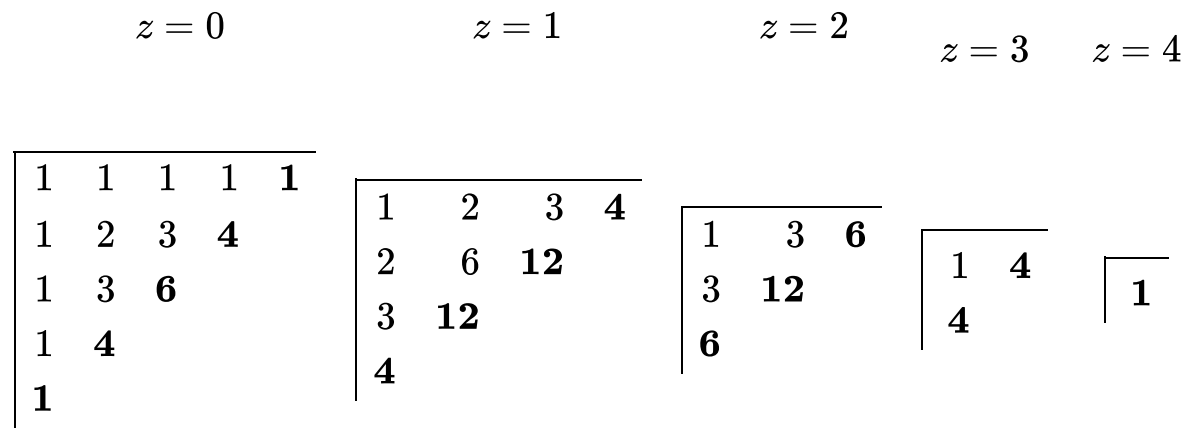
1	1	1	1	1
1	2	3	4	
1	3	6		
1	4			
1				

Pascal's hyper-pyramids generalize the triangle to higher dimensions. In 3 dimensions, the value at position (x, y, z) is the sum of up to three other values:

- $(x, y, z - 1)$, the value immediately below it if we are not on the bottom face ($z = 0$);
- $(x, y - 1, z)$, the value immediately behind it if we are not on the back face ($y = 0$);
- $(x - 1, y, z)$, the value immediately to the left if we are not on the leftmost face ($x = 0$).

All values at the bottom, back, and leftmost faces are 1.

The following figure depicts Pascal's 3D-pyramid of height 5 as a series of plane cuts obtained by fixing the value of the z coordinate.



For example, the number at position $x = 1, y = 2, z = 1$ is the sum of the values at $(0, 2, 1), (1, 1, 1)$ and $(1, 2, 0)$, namely, $6 + 3 + 3 = 12$. The base of the pyramid corresponds to a plane of positions such that $x + y + z = 4$ (highlighted in bold above).

The size of each layer grows quadratically with the height of the pyramid, but there are many repeated values due to symmetries: numbers at positions that are permutations of one another must be equal. For example, the numbers at positions $(0, 1, 2), (1, 2, 0)$ and $(2, 1, 0)$ above are all equal to 3.

Task

Write a program that, given the number of dimensions D of the hyper-space and the height H of a hyper-pyramid, computes the set of numbers at the base.

Input

A single line with two positive integers: the number of dimensions, D , and the height of the hyper-pyramid, H .

Constraints

$2 \leq D < 32$ Number of dimensions

$1 \leq H < 32$ Height

D and H are such that all numbers in the hyper-pyramid are less than 2^{63} .

Output

The set of numbers at the base of the hyper-pyramid, with no repetitions, one number per line, and in ascending order.

Sample Input 1

2 5

Sample Output 1

1
4
6

Sample Input 2

3 5

Sample Output 2

1
4
6
12

